Joshua Osborne

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Summary

Physics PhD with a GNN classifier at 0.97 ROC-AUC and 6 peer-reviewed publications in Bayesian inference and statistical modeling. In industry, delivered a 10x ROAS improvement through marketing attribution analysis and built production AWS pipelines adopted across sales, marketing, and leadership.

Experience

RipDrip — **December 2025 – April 2026**

Data Scientist — *Arlington, TX*

- Delivered a 10x increase in ad-spend ROAS by identifying campaign inefficiencies via SQL attribution analysis in AWS Athena; findings presented directly to leadership.
- Defined and delivered key sales KPIs (show-rate, time-to-sale, campaign efficiency) in a Power BI dashboard with daily automated refresh, replacing a fully manual process; adopted by sales and marketing as the primary reporting layer for weekly performance reviews.
- Eliminated manual review of 1,000+ daily messages by building a Python pipeline combining regex and LLM inference (Gemma 4, few-shot + chain-of-thought) to extract and structure customer intent signals for automated sales prioritization.
- Architected an end-to-end ETL pipeline on AWS (PySpark via Docker for local testing, Glue, S3, Athena) ingesting from 6 APIs; medallion architecture with schema enforcement in silver, star schema in gold; configured SNS alerting for pipeline failure detection and data health monitoring; enforced least-privilege access via IAM policies and KMS encryption across S3 and Glue resources.

University of Texas at Arlington — **January 2020 – December 2025**

Graduate Research Assistant — *Arlington, TX*

- Published 6 peer-reviewed papers applying Bayesian inference, MLE, Monte Carlo simulation, bootstrap methods, and hypothesis testing in Python to large astrophysical datasets.
- Resolved contested empirical claims across multiple peer-reviewed publications by applying significance testing and root cause analysis to identify how model misspecification propagated into false conclusions.
- Built and evaluated deep learning models using PyTorch across 6 GPUs; operated MPI-parallelized scientific computing pipelines on TACC supercomputers (96 cores).
- Presented research findings to both technical and non-technical audiences at multiple American Physical Society (APS) conferences.
- Mentored 3 undergraduate researchers in scientific computing, statistical analysis, and Python.

Projects

Graph Neural Network Classifier for High-Energy Physics Event Detection

- Achieved **0.97 ROC-AUC** on a balanced dataset, outperforming an XGBoost baseline, by building a GNN classifier on 500K 3D point cloud files (300 GB) for graph-level signal vs. background classification.
- Engineered radius-based and k-NN graph topologies (`torch-cluster`), trained across 6 GPUs with early stopping; optimized classification threshold via Youden's J statistic, evaluated via confusion matrix, precision/recall, and ROC curve.
- Generalizes to any binary classification problem on high-dimensional, structured/relational data (e.g. fraud detection) where raw graph structure outperforms hand-engineered features.

Urban Safety Atlas: County-Level Risk Index (3,144 US Counties)

- Validated composite county safety index against life expectancy (Spearman $\rho = 0.77$, Pearson $r = 0.74$) and three secondary benchmarks across 3,144 US counties; all directional checks passed without the benchmark variables ever entering the model.
- Quantified geographic clustering (Global Moran's I = 0.627, p < 0.001); KMeans and HDBSCAN identified distinct regional risk tiers; Kruskal-Wallis confirmed significant disparities (H = 805.7, p < 0.001) across US regions.
- Built index by merging 4 government sources (EPA, CDC PLACES, NHTSA FARS, Census SAIPE); applied IDW imputation for 2,170 counties lacking EPA monitors and empirical Bayes shrinkage to the traffic fatality rate to prevent small-population counties from dominating the scale; deployed Streamlit dashboard with interactive sub-score weight sliders and live cluster re-fitting.

Hierarchical Bayesian Inference for Population-Level Estimation (1,366 objects)

- Constrained redshift (cosmological distance proxy) estimates to 0.71–1.40 uncertainty at 90% confidence across 4 competing model specifications by fitting a 16-parameter posterior via adaptive MCMC on 96 cores.
- Addressed missing data across a 5-D incomplete dataset (1,366 objects) using fuzzy c-means clustering to identify structural groupings, then designed a multilevel hierarchical Bayesian model via Empirical Bayes / EM with detection-bias corrections to infer the missing values.
- Published framework for population-level Bayesian inference with missing data and detection thresholds; directly transferable to any domain requiring principled imputation at scale.

Statistical Modeling & Model Selection for Large Observational Datasets

- Overturned a widely cited result, showing it was a statistical artifact; BIC model comparison favored the null model by a factor of 10^6–10^9, validated across 50K Monte Carlo simulations.
- Quantified how selection bias, sample incompleteness, and binning distort inferred parameters by fitting lognormal and Pareto distributions via MLE (SciPy) to a catalog of $\sim$3,000 astrophysical events.
- Methodology applicable to any domain where data processing choices introduce spurious signals; published in *Astronomy & Astrophysics*.

Interactive Statistics & ML Reference Site | joshuaosbornedata.github.io/notes

- Built a public statistics and machine learning reference site (ongoing) with a custom Python build pipeline including Markdown preprocessing, HTML templating, and static site generation.
- Implemented Thompson Sampling (Beta-Bernoulli conjugate priors), UCB1, and ε-greedy from scratch; benchmarked cumulative regret across 200 runs $\times$ 10,000 rounds; results rendered as original animated visualizations in Matplotlib.
- Built a CUPED variance-reduction simulation over 2,000 replicate A/B experiments; empirical variance ratio (0.36) matched theoretical $1-\rho^2$ (0.37), demonstrating a $\sim$64% cut in required sample size at $\rho = 0.8$.

Technical Skills

Languages: Python 3, SQL, PostgreSQL, Julia, MATLAB
Machine Learning (ML) & Statistics: PyTorch, Scikit-learn, XGBoost, SciPy, Statsmodels; graph neural networks (PyTorch Geometric, torch-cluster); Bayesian inference, MLE, GLMs, regression analysis, Monte Carlo simulation, bootstrap methods, clustering (k-means, DBSCAN, HDBSCAN), spatial statistics (IDW imputation, Moran's I), time series analysis
Experimentation: A/B testing, CUPED variance reduction, multi-armed bandits (Thompson Sampling, UCB1, ε-greedy), SRM detection, sequential testing / alpha spending, power analysis
Data & Viz: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit, Power BI, Tableau
Cloud (AWS): S3, Glue, Athena, Lambda, Eventbridge, IAM, KMS
Tools: PySpark, Conda, Git, Docker, Linux, Jupyter, web scraping

Awards/Achievements

Graduate Assistance in Areas of National Need (GAANN) Fellowship — Spring 2025 – Fall 2025

- Awarded by the U.S. Department of Education and UTA to support graduate research in fields of national interest.

Argonne Training Program on Extreme-Scale Computing (ATPESC) — Summer 2024

- Selected among $\sim$70 participants worldwide for intensive training in scalable computing methods directly applicable to large-scale data pipelines.

Student Employee of the Year — Spring 2021

- Awarded Student Employee of the Year by the Department of Physics at UTA in recognition of exceptional performance as a graduate research and teaching assistant.

Education

University of Texas at Arlington — **December 2025**
Ph.D. in Physics — *Arlington, Tx*

University of Texas at Arlington — **December 2019**
B.Sc. in Physics with a minor in Mathematics — *Arlington, Tx*